

Entrepreneurship (MG4011)

Final Exam

Date: May 18th, 2026

Total Time (Hrs): 3

Course Instructor

Total Marks: 100

Mr. Malik Zaeem Ul Haq

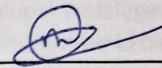
Total Questions: 4

221-6790

Roll No

✓ BCS-8A & 8B

Section


Student Signature

Instructions/Notes

- Clearly mention your Roll No. & Section on both question and answer paper;
- Use **blue** or **black** pen only;
- Use of **Calculator** is allowed;
- Record answers of MCQs on answer booklet e.g. 1A-2B-3C-4D etc.;
- **Answers to all the questions must be in the answer booklet to be submitted;**
- See through all Sections once before starting and closely follow all instructions given.

Attempt all the questions

Q1 - MCQs [1 mark each, Total = 40]

1. A software startup reports a net profit of Rs 500,000 for the quarter but files for bankruptcy the following month. Based on startup financial metrics, what is the most likely cause?

- A) Their contribution margin ratio (CMR) was too high.
- B) Their CAC payback period was under 6 months.
- ☒ C) They had positive net income but negative cash flow due to high accounts receivable and high burn rate.
- D) Their LTV:CAC ratio exceeded the 3:1 benchmark.

2. A SaaS company has an LTV:CAC ratio of 8:1. While this sounds excellent, what strategic error does this high ratio most likely indicate?

- A) The company is over-spending on customer acquisition and burning cash too fast.
- ☒ B) The company is severely under-investing in marketing and missing out on rapid market capture.
- C) The company's churn rate is dangerously high.
- D) The company has priced its product too low for the market.

3. In cohort retention analysis, what does a "smile curve" indicate about a startup's user base?

- A) Users are dropping off immediately after the first day of use.
- B) The CAC is steadily decreasing over time.
- C) Churn is accelerating month over month.
- ☒ D) Users who previously stopped using the app are returning, indicating strong long-term Product-Market Fit.

4. A startup has a monthly ARPU of Rs 200, a Gross Margin of 75%, and a monthly churn rate of 5%. What is the Customer Lifetime Value (CLV)?

- A) Rs 3,000
- ☒ B) Rs 4,000
- C) Rs 15,000
- D) Rs 750

$$CLV = \frac{ARPU}{Churn} = 4000$$

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5. A digital health platform has a CAC of Rs 1,200. The monthly ARPU is Rs 200, but the Gross Margin is 50% due to API and cloud hosting costs. What is the CAC Payback Period?

- A) 3 months
- B) 6 months
- ☒ C) 12 months
- D) 24 months

$$\text{CAC pay} = \frac{\text{CAC}}{\text{200 ARPU Contribution}} = \frac{1200}{200 \times 100} = 6$$

$$= 12$$

6. Analyzing the Airlift case study, what was the primary structural reason the company failed despite massive funding and scaling?

- A) They lacked a functioning Minimum Viable Product (MVP).
- B) They ignored local regulatory requirements in Pakistan.
- C) The founders refused to pivot from transit to quick commerce.
- ☒ D) They scaled prematurely with broken unit economics, leading to an unsustainable burn rate.

7. A startup has cash reserves of Rs 5,000,000. Their monthly revenue is Rs 1,000,000 and total monthly expenses are Rs 2,500,000. What is their net burn rate and runway?

- A) Net burn: Rs 2,500,000; Runway: 2 months
- ☒ B) Net burn: Rs 1,500,000; Runway: 3.33 months
- C) Net burn: Rs 1,000,000; Runway: 5 months
- D) Net burn: Rs 1,500,000; Runway: 5 months

$$\text{Burn} = 1,000,000 - 2,500,000 = -1,500,000$$

$$\text{Runway} = \frac{\text{cash}}{\text{burn}} = \frac{5,000,000}{1,500,000} = 3.33$$

8. A startup has LTV:CAC = 4:1, CAC payback = 20 months, churn = 3% monthly. Best decision?

- A) Scale marketing aggressively
- ☒ B) Improve onboarding and retention first
- C) Increase prices immediately
- D) Raise more funding to extend runway

↓
a lot

9. An app has 10,000 current users. On average, each user sends 3 invites to friends, and 20% of those invites convert into new active users. What is the K-factor, and what does it signify?

- A) K = 0.6; The app is growing exponentially through virality.
- B) K = 1.2; The app is experiencing sub-viral, declining growth.
- ☒ C) K = 0.6; The app is sub-viral and will eventually stop growing organically without paid marketing.
- D) K = 6.0; The app has achieved perfect Product-Market Fit.

$$K = i \times \frac{20}{100} \times 3$$

$$= 3 \times 0.6 = 1.8$$

10. App X has 50,000 MAU and 25,000 DAU. App Y has 100,000 MAU and 10,000 DAU. Which app has better "stickiness" and why?

- ☒ A) App X, because its DAU/MAU ratio is 50%, indicating high daily habit formation.
- B) App Y, because it has double the total Monthly Active Users.
- C) App Y, because a 10% DAU/MAU ratio is the optimal benchmark for tech startups.
- D) App X, because a lower absolute MAU makes server costs cheaper.

$$\frac{\text{DAU}}{\text{MAU}} = \frac{25,000}{50,000} = \frac{1}{2} = 0.5$$

$$\frac{10,000}{100,000} = \frac{1}{10} = 0.1$$

11. During an elevator pitch, a founder successfully covers the problem, the solution, the target market, and the unique value proposition. However, investors pass on the idea. Based on the course framework, which critical element was omitted?

- ☒ A) The exit strategy for the investors.
- B) A detailed 5-year financial balance sheet.
- C) The exact lines of code used in the MVP.
- D) The monetization strategy (how the startup captures value).

12. Why is a high churn rate fundamentally more dangerous to a SaaS startup than a low conversion rate?

- A) Low conversion rates immediately crash the server infrastructure.
- ☒ B) A high churn rate destroys CLV and acts like a "leaky bucket," rendering customer acquisition spending useless over time.
- C) High churn reduces the Gross Margin percentage directly.
- D) High churn artificially inflates the DAU/MAU ratio.

13. In the context of the startup lifecycle, which of the following is the most accurate definition of a "Pivot"?

- ☒ A) Scrapping the entire business model and starting a completely unrelated company.
- ☐ B) Fixing minor bugs in the software based on user feedback.
- ☐ C) Changing one or more fundamental assumptions of the business model while keeping the core vision intact.
- ☐ D) Firing the technical team to reduce the monthly burn rate.

14. A startup uses a Freemium model. They have 100,000 free users and 2,000 paid users. The company is running out of money. What is the fundamental unit economics failure here?

- ☒ A) The conversion rate is too high, leading to server crashes.
- ☐ B) The cost to serve the free users (COGS) exceeds the contribution margin generated by the paid users.
- ☐ C) The ARPU of the free users is negatively impacting the K-factor.
- ☐ D) Freemium models inherently cannot achieve a CMR above 0.20.

15. The "Fainthearted" case study primarily illustrates which core aspect of entrepreneurship?

- ☐ A) Writing perfect business plans guarantees venture capital funding.
- ☐ B) Virality is the only way to scale a consumer application.
- ☐ C) Bootstrapping is always superior to raising venture capital.
- ☒ D) Entrepreneurial decision-making involves managing severe risk, uncertainty, and personal resilience.

16. A two-sided marketplace app connects freelance developers with clients. The platform becomes exponentially more valuable to clients only when more developers join, and vice versa. This is an example of:

- ☒ A) Direct Network Effects.
- ☐ B) Economies of Scale.
- ☐ C) Indirect Network Effects.
- ☐ D) The Freemium Paradox.

17. A SaaS product is priced at Rs 1,000/month. The variable cost per user (AWS, API calls) is Rs 400. Fixed costs are Rs 600,000/month. What is the Contribution Margin Ratio (CMR) and the break-even volume?

- ☐ A) CMR = 0.40; Break-even = 1,500 users
- ☐ B) CMR = 0.60; Break-even = 1,500 users
- ☒ C) CMR = 0.60; Break-even = 1,000 users
- ☐ D) CMR = 0.40; Break-even = 1,000 users

$$\text{contri} = 1000 - 400 = 600 \quad \text{CMR} = \frac{600}{1000} = 0.6$$

$$\text{BE} = \frac{600,000}{600} = 1000$$

18. According to the course foundations, what is the primary distinction between a Startup and a Small Business?

- ☒ A) Startups are designed to search for a repeatable, highly scalable business model, while small businesses intend to operate stably within a known model.
- ☐ B) Startups only sell digital software, whereas small businesses sell physical goods.
- ☐ C) Startups must raise venture capital, while small businesses rely strictly on bank loans.
- ☐ D) Small businesses focus on CDC Value, while startups focus only on capturing value.

create
distribute
capture

19. A technical founder proves that their new database architecture solves data lag (Problem-Solution Fit). However, after launch, no one pays for it. What stage did the startup fail to achieve?

- ☒ A) Minimum Viable Product (MVP)
- ☐ B) Product-Market Fit (PMF)
- ☐ C) Viral Coefficient Optimization
- ☐ D) Ideation Phase

20. A mobile app generates Rs 500,000 in monthly recurring revenue from a total base of 10,000 active users. What is the ARPU?

- ☐ A) Rs 0.02

$$\frac{500,000}{10,000} = 50$$

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CLV > CAC good.

- B) Rs 5
C) Rs 50
D) Rs 500

21. A B2B software company discovers its CAC is Rs 10,000 and its CLV is Rs 8,000. What is the harsh reality of this metric combination?
- A) The company is highly profitable and should scale marketing immediately.
B) The company has achieved the Golden Ratio.
C) The company's payback period is less than one month.
D) The business model is fundamentally broken; they lose Rs 2,000 on every customer they acquire.
22. An opportunity must pass 5 filters: Problem Severity, Market Size, Economics, Timing, and ...?
- A) Patentability
B) Venture Capital Interest
C) Feasibility (Can the team actually build and deliver it?)
D) Social Impact
23. A consumer app has a K-factor of 1.5, yet the startup's cash reserves are draining rapidly and revenue is flat. How is this mathematically possible?
- A) The virality is bringing in non-paying users, but the high churn rate or server costs (COGS) are outpacing the value of new users.
B) A K-factor of 1.5 means the user base is shrinking by 50% every month.
C) The founders did not calculate the DAU/MAU ratio correctly.
D) The LTV:CAC ratio is 10:1, which drains cash automatically.
24. In the context of early-stage entrepreneurship, which statement best defines the difference between an "Idea" and an "Opportunity"?
- A) An idea has paying customers, while an opportunity is just a thought.
B) An idea requires a tech stack, while an opportunity requires marketing.
C) An opportunity only exists in B2B markets, while ideas are for B2C markets.
D) An idea is a concept, while an opportunity is an idea backed by market demand, feasibility, and viable unit economics.
25. A startup pushes aggressive Facebook ad campaigns before fixing widespread bugs that cause users to delete the app within 2 days. Which startup anti-pattern does this represent?
- A) The Fast-Follower Strategy
B) Premature Scaling
C) Effectuation
D) Lean Development
26. In software unit economics, what does a very high Gross Margin (e.g., 85%) signify?
- A) The startup has very low variable costs (like server hosting and APIs) relative to its revenue, making it highly scalable.
B) The startup is spending too much money on marketing and sales.
C) The startup has a dangerously high burn rate.
D) The startup's fixed costs are too high to achieve break-even.
27. A landing page receives 20,000 visitors. 500 of them sign up for a free trial. What is the conversion rate?

$$2.5 = \frac{5}{2} = \frac{500}{20,000} \times 100 = \frac{20,000}{500} = 40$$

- ☒ C) 2.5%
D) 0.025%

28. In the "CDC Value" framework (Create, Deliver, Capture), establishing a highly profitable subscription pricing tier represents which phase?

- A) Create
B) Consolidate
C) Deliver
☒ D) Capture

29. What is the fundamental purpose of building a Minimum Viable Product (MVP)?

- ☒ A) To maximize validated learning about customers with the least amount of effort and resources.
B) To launch a bug-free, fully-featured software application to beat competitors.
C) To secure a bank loan by showing a completed physical asset.
D) To immediately achieve a positive net income.

Soft as a solution.

30. Comparing Airlift to successful digital models, why is a pure software SaaS generally easier to scale globally than a quick-commerce delivery platform?

- A) SaaS products do not require any marketing expenditure.
☒ B) SaaS has negligible marginal delivery costs and no physical logistics constraints, whereas quick-commerce is bound by physical assets and local delivery COGS.
C) Quick-commerce models suffer from direct network effects, slowing them down.
D) SaaS products always have a lower churn rate than physical delivery services.

31. A startup decides to increase its software subscription price from Rs 1,000 to Rs 1,200, while keeping variable costs the same. How does this affect the unit economics?

- ~~A) It decreases the Contribution Margin Ratio (CMR) and increases Break-even units.~~
B) It has no effect on Break-even units because fixed costs didn't change.
☒ C) It increases the Contribution Margin Ratio (CMR) and decreases the Break-even units.
D) It increases the Gross Margin but destroys the LTV:CAC ratio.

32. A startup has an excellent LTV:CAC ratio of 5:1 and a high DAU/MAU ratio. Yet, they are projected to run out of cash in 3 months. What is the most logical explanation?

- ~~A) Their K-factor has dropped below 1.0.~~
~~B) They have negative gross margins.~~
C) Their conversion rate is too high, leading to server overload.
☒ D) They have massive fixed costs (e.g., huge R&D payroll or office leases) that exceed their total contribution margin, causing a high net burn.

33. What does "Net Negative Churn" mean in the context of SaaS metrics?

- ☒ A) The revenue gained from existing customers upgrading their plans exceeds the revenue lost from customers who cancel.
~~B) Zero customers have cancelled their subscriptions in the current month.~~
C) The total number of users is shrinking, but CAC is also shrinking.
D) The company is losing money on every single cohort.

34. A founder claims they have "validated the market" because 5,000 people liked their Facebook page and answered "yes" to a survey about buying the product. Why is this dangerous from a data-driven perspective?

- A) Facebook is no longer a valid marketing channel.

35. Which of the following CAC Payback Periods is generally considered the healthiest benchmark for a fast-growing tech startup?
- ☒ A) 6 to 12 months
 - ☐ B) 24 to 36 months
 - ☐ C) 48 months
 - ☐ D) Payback period doesn't matter if CLV is positive.

36. A student developer confuses "Gross Revenue" with "Gross Margin" when calculating LTV. What will be the strategic consequence of this math error?

- A) They will underestimate their CAC payback period.
- B) They will accidentally lower their DAU/MAU ratio.
- C) They will significantly overestimate the lifetime value of a customer, potentially leading to dangerous overspending on customer acquisition.
- D) They will falsely believe their fixed costs are too high.

37. An app currently has 5,000 active users. The K-factor is calculated at 1.2. If a single referral cycle completes, how many new users will be generated, and is this growth sustainable?

- A) 5,000 new users; No, it is not sustainable.
B) 1,200 new users; Yes, it is viral.
C) 12,000 new users; No, it is sub-viral.
D) 6,000 new users; Yes, it is viral and exponential.

$$5000 \times 1.2 \quad K = 1.2$$

38. Which statement best summarizes why course assessments integrate multiple metrics (e.g., combining Burn Rate, LTV:CAC, and Churn) into single scenarios?

- ☐ A) To make the math unnecessarily difficult for non-finance students.
- ☒ B) Because in reality, optimizing one metric (like CAC) often negatively impacts another (like Churn or Burn), and entrepreneurs must balance the entire system.
- ☐ C) Because investors only care about the K-factor.
- ☐ D) To prove that traditional accounting (Net Income) is useless.

39. A startup's dashboard shows: Burn Rate = Rs 1,000,000/month, Runway = 2 months, LTV:CAC = 1:2. The founder wants to hire 5 new developers to build more features. What is the correct data-driven decision?

- ☒ A) Do not hire. Cut costs immediately to extend the runway, and focus on fixing the broken LTV:CAC ratio (unit economics) before building anything new.
- ☐ B) Hire the developers. New features will automatically increase the LTV:CAC ratio.
- ☒ C) Double the marketing budget to fix the runway.
- ☐ D) Pivot to a quick-commerce delivery model.

40. Ultimately, what is the defining trait of an entrepreneur that the course aims to develop, particularly highlighted in case studies like Fainthearted?

- A) The ability to write code faster than competitors.
B) The ability to raise millions in venture capital without a product.
C) The capacity to memorize financial definitions perfectly.
D) The ability to make critical, data-driven decisions under extreme pressure, risk, and uncertainty.

Q2 - Numericals [4 x 5 = 20 marks]

1. A startup sells a product at Rs 3,000 per unit. Variable cost per unit is Rs 1,800. Fixed costs are Rs 1,200,000 per month.

Calculate:

- a) CMR
- b) Break-even units
- c) Break-even revenue

2. A startup has **CAC = Rs 2,400**, monthly contribution per customer = **Rs 600**, and average retention = **6 months**.

Calculate:

- a) CLV
- b) LTV:CAC ratio
- c) CAC payback period
- d) Give a one-line comment on whether the model is healthy

3. A startup has Revenue = **Rs 1,000,000**. Expenses = **Rs 1,200,000**, including depreciation of **Rs 300,000** which is non-cash.

Calculate:

- a) Net income
- b) Burn rate
- c) Is the startup profitable yet still burning cash? Explain.

4. A digital startup has **40,000 visitors**, **1,600 buyers**, **12,000 MAU**, **2,400 DAU**, and monthly revenue of **Rs 300,000**.

Calculate:

- a) Conversion rate
- b) DAU/MAU ratio
- c) ARPU

Q3 - Short Analysis Questions [20 marks]

1. (4 marks)

What are the financial objectives of a firm? Mention any four and briefly explain.

2. (4 marks)

What are the 4 Ps of marketing? Explain each with one example.

3. (3 marks)

Differentiate between **Pivoting** and **Scaling** with startup context.

4. (3 marks)

Why can a startup with **LTV:CAC = 5:1** still be making a strategic mistake?

5. (3 marks)

Explain why **high churn** is more dangerous than **low acquisition**.

6. (3 marks)

A startup has **DAU/MAU = 15%**. What does this indicate about Product-Market Fit?



Profit max
wealth max
survival
Business g

Price
Prod
Prom
Place

Q4 - Mini Case: CampusLoop

CampusLoop is a Lahore-based freemium app for university students. It helps users manage assignments, share notes, and form study groups. The startup has grown to 24,000 MAU, 3,600 DAU, and 900 paid users. Monthly revenue is Rs 450,000. Customer acquisition cost is Rs 1,800 per paid customer, monthly contribution per paid user is Rs 300, monthly churn is 21%, cash in bank is Rs 1,500,000, and monthly burn is Rs 420,000. The founders want to expand to three more universities immediately. There is no formal founder agreement yet, and one founder suggests showing projected DAU as current figures in the investor pitch.

Questions [4 x 5 = 20 marks]

- a) What business model is CampusLoop using? Mention one strength and one weakness of this model.
- b) Based on the given metrics, should the founders scale now? Give two reasons.
- c) Identify one legal issue and one ethical issue in the case.
- d) Suggest two immediate corrective actions before any expansion.